

Markscheme

Mathematics: analysis and approaches
Practice question bank

49. (a)

Consider the equation $3^{2x-1} = 5^{x+2}$.

Solve the equation for x , giving your answer correct to 3 significant figures.

$$\text{take } \ln \text{ of both sides: } (2x - 1) \ln 3 = (x + 2) \ln 5 \quad (M1)$$

$$2x \ln 3 - \ln 3 = x \ln 5 + 2 \ln 5 \quad \mathbf{A1}$$

$$x(2 \ln 3 - \ln 5) = 2 \ln 5 + \ln 3 \quad (M1)$$

$$x = \frac{2 \ln 5 + \ln 3}{2 \ln 3 - \ln 5} \quad \mathbf{A1}$$

$$x = 7.35 \quad \mathbf{A1}$$

[5 marks]